

Reaction of Vinyl Ethers with ArSCl Adducts of D-Glucal

Irina P. Smoliakova^{a*}, Mingming Han^a, Jianchun Gong^b, Ron Caple^b, William A. Smit^c

^a Chemistry Department, University of North Dakota, Grand Forks, ND 58202-9024, USA

^b Chemistry Department, University of Minnesota-Duluth, 10 University Drive, MN 55812, USA

^c Zelinsky Institute of Organic Chemistry, pr. Lenina, 47, Moscow, 117913, Russia

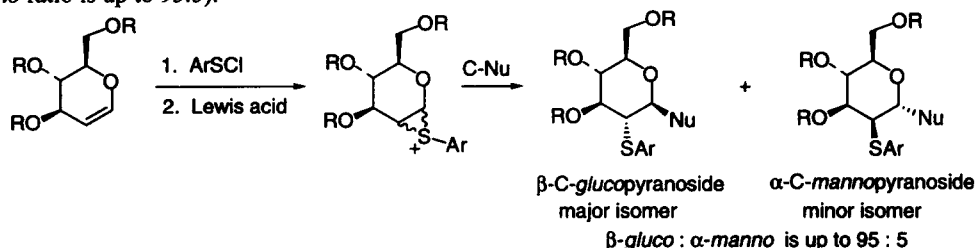
Received 18 December 1998; revised 11 February 1999; accepted 16 February 1999

Abstract: Lewis acid mediated reactions of vinyl ethers with ArSCl adducts of benzyl protected D-glucal followed by quenching of the five-membered sulfonium salt intermediates with external nucleophiles, H₂O, MeOH, and *n*-Bu₄NBH₄, lead to a highly stereoselective formation of β -C-glucopyranosides with different functional groups in the lateral chain. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: C-glycosides, glycals, vinyl ethers, sulfonium salts.

Introduction

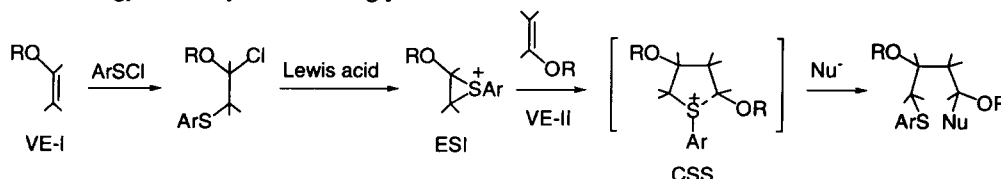
During the past two decades the synthesis of C-glycosides has been an intriguing topic in both carbohydrate and organic chemistry.¹ Compounds of this type have been found in nature and often possess a variety of physiological activities.² As non-hydrolyzable analogs of O-glycosides, carbon-linked glycosides have been used as enzyme inhibitors^{3,4} and substrates for carbohydrate-binding proteins.³ Numerous methods have been proposed for the synthesis of C-glycosides, but the majority of these procedures give mixtures of α - and β -anomers with predominant formation of α -C-glycosides.^{1,5,6} Recently we have reported that, in the presence of a Lewis acid, ArSCl adducts of tri-O-benzyl-D-glucal are capable of reacting with a number of C-nucleophiles including TMSCN, silyl enol ethers, silyl ketene acetals, allylsilanes, and Grignard reagents (Scheme 1).⁷ These reactions provide β -C-glucopyranosides with high stereoselectivity (the β -gluco : α -manno ratio is up to 95:5).



Scheme 1.

* E-mail: ismoliakova@mail.chem.und.nodak.edu

Vinyl ethers have not been used in organic synthesis as broadly as silyl enol ethers or other enols. To the best of our knowledge, vinyl ethers have not been used as carbon nucleophiles in the synthesis of *C*-glycosides and related compounds. Previously we reported the ArSCI mediated condensation of two vinyl ethers, in which the first vinyl ether was used as a precursor of a *C*-electrophile (an episulfonium ion) and the second vinyl ether was employed as a *C*-nucleophile.⁸ The reaction proceeds through the formation of two intermediates, the episulfonium ion and the cyclic sulfonium salt (CSS). The later intermediate was isolated and its structure was established by X-ray crystallographic analysis.^{8g} The cyclic sulfonium salt is capable of reacting with various nucleophiles (Scheme 2). Here we would like to report the application of this methodology for the synthesis of *C*-glycosides.

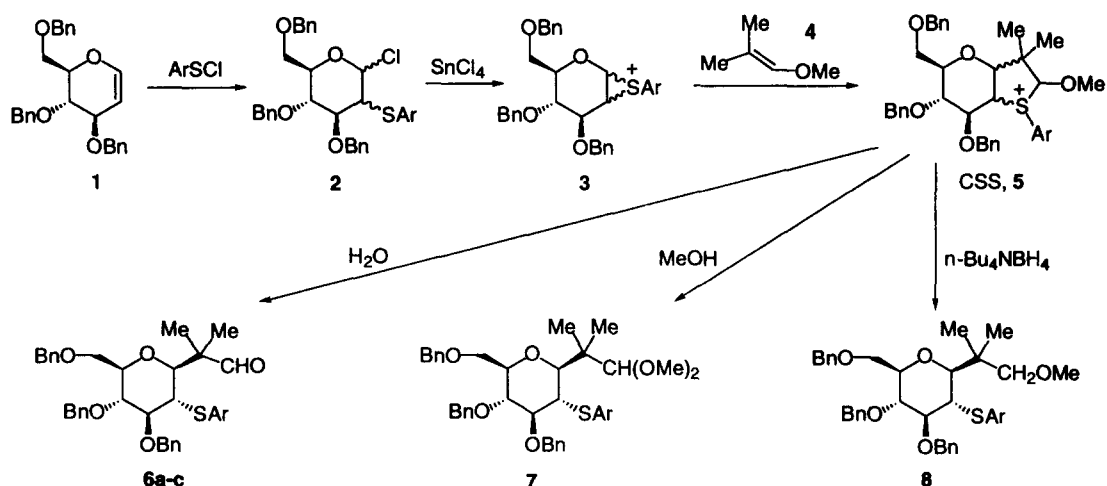


Scheme 2.

Results and Discussion

The commercially available tri-*O*-benzyl-D-glucal (**1**) has been chosen as a model glycal for the present study. Reaction of **1** with *p*-TolSCI was carried out as described previously.^{7a} The products of this reaction (CH_2Cl_2 , room temperature) are three isomeric chlorides (**2**), β -*gluco*-2, α -*gluco*-2, and α -*manno*-2, in a ratio of 88:12 (*gluco* : *manno*).^{7a} It was found that, in the presence of SnCl_4 , the reaction of chloride mixture **2** (prepared *in situ* in CH_2Cl_2 at room temperature) with 1-methoxy-2-methyl-1-propene (**4**) at -78°C in CH_2Cl_2 followed by quenching the reaction mixture with water resulted in the formation of aldehyde **6a** as a mixture of β -*gluco* and α -*manno* isomers in a ratio of 87:13 in a combined yield of 87% (Scheme 3). The high yield and the absence of products of vinyl ether oligomerization suggest to us that the reaction is likely to take place through the formation of the stabilized intermediate, the cyclic sulfonium salt **5**, as it was observed in the ArSCI mediated dimerization of vinyl ethers.^{8,9} To support this, we quenched the reaction mixture with other nucleophiles, MeOH and *n*- Bu_4NBH_4 .¹⁰ The use of MeOH led to the formation of acetal **7**, while treatment of the reaction mixture with *n*- Bu_4NBH_4 gave rise to reduced product **8**.

In order to determine the scope and limitations of the reaction, other vinyl ethers have also been employed. Thus, the reaction of adduct mixture **2** with methyl vinyl ether after quenching with water provides a mixture of two isomeric aldehydes **11** (β -*gluco* : α -*manno* ratio is 91:9) in a combined yield of 63%. The use of MeOH and *n*- Bu_4NBH_4 instead of H_2O as the final nucleophile in the same reaction provided the corresponding *C*-glycosides having acetal and ether functional groups in the lateral chain (see Table 1). Aldehyde **11** has also been synthesized in the same yield by the reaction of chloride mixture **2** with ethyl vinyl ether. The use of the bulkier nucleophile, propyl vinyl ether, led to the formation of the same *C*-glycoside but in the low yield of 18%. Our numerous attempts to obtain aldehyde **11** using *n*-butyl vinyl ether



β -*gluco* : α -*manno* ratios are up to 99:1 (see Tables 1, 2, and 3).

Scheme 3

were unsuccessful. The main product of the latter reaction was 3,4,6-tri-*O*-benzyl-2-(*p*-tolylsulfanyl)-*D*-glucose although some other unidentified compounds have been isolated as well. One could suggest that in the case of a rather bulky alkoxy group, such as *n*-butyl, the five-membered sulfonium salt intermediate is very unstable or can not be formed at all. However, the reaction of ethylene glycol methyl vinyl ether with adduct mixture **2** provided two isomers of *C*-glycoside **11** in a good yield (see Table 1).

The use of 2-methoxypropene, an α -substituted vinyl ether, as a *C*-nucleophile did not provide the desired *C*-glycoside. However, another vinyl ether with the bulky electron-releasing trimethylsilyl group in the α -position (**17**) smoothly reacted with adducts **2a–c** to give *C*-glycoside **18** in a yield of 59%.

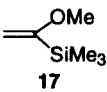
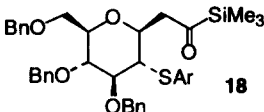
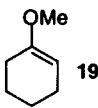
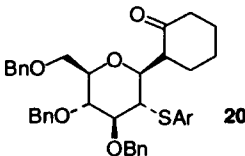
The reactions with two cyclic vinyl ethers, 1-methoxycyclohexene (**19**) and 1-methoxycyclopentene, gave contrary results. The use of **19** provided a mixture of only two isomeric *C*-glycosides (**20**, 10:1) in a good yield (61%). In the contrast to other vinyl ethers, these two isomeric compounds both have β -*gluco* configuration (¹H NMR data) and differ in the configuration of the chiral center in the lateral chain. The (*S*)-configuration of the chiral center of the major isomer **20** has been established using a single crystal crystallographic analysis of the corresponding sulfone. The reaction with 1-methoxycyclopentene did not provide any traces of the expected *C*-glycoside. An examination of molecular models of the required sulfonium salt intermediate for this reaction indicates a significant angle strain.

We also studied the stereoselectivity of the last step of the reaction sequence: the reaction of a final nucleophile with the second intermediate, the five-membered sulfonium salt. To examine stereoselectivity, *n*-Bu₄NBD₄ was used as the final nucleophile. The product of this reaction was a mixture of two isomeric *C*-glycosides (**9**) (β -*gluco* : α -*manno* = 95:5). The main β -*gluco* isomer was isolated as a mixture of two diastereomers (ca. 1:1) which differ in the configuration at the chiral center in the lateral chain (NMR data).

Table 1. Reaction of p-TolSCl Adducts of Glucal 1 with Vinyl Ethers.^a

Entry	VE	Nu	Product	Yield, % ^b	Ratio β-gluco : α-manno
1		H ₂ O		87	87:13
2	4	MeOH		76	87:13
3	4	n-Bu ₄ NBH ₄		84	87:13
4	4	n-Bu ₄ NBD ₄		58	92:8
5		H ₂ O		63	91:9
6	10	MeOH		34	91:9
7	10	n-Bu ₄ NBH ₄		23	91:9
8		H ₂ O	10	63	95:5
9		H ₂ O	10	18	88:12
10		H ₂ O	10	42	91:9

Table 1 - Continuation

Entry	VE	Nu	Product	Yield, %	Ratio β -gluco : α -manno
11		H ₂ O		59	94:6
12		H ₂ O		61	90:10

^a All reactions were carried out in CH₂Cl₂ in the presence of 1.2 equiv of SnCl₄ at -78 °C.

^b Combined yield of β -gluco and α -manno isomers

Table 2. Effect of Lewis Acid on the Stereoselectivity of the Reaction of p-TolSCl Adducts of Glucal 1 with Vinyl Ether 4.

Entry	Lewis Acid	β -gluco-6a : α -manno-6a ratio	Yield, %
1	SnCl ₄	87:13	87
2	TiCl ₄	93:7	32
3	BF ₃ ·OEt ₂	87:13	93
4	ZnCl ₂	95:5	95

Table 3. Effect of ArS Substituent on the Stereochemistry of the Reaction of ArSCl Adducts of Glucal 1 with Vinyl Ether 4.

Entry	ArS Group	β -gluco-6 : α -manno-6 ratio	Yield, %
1	p-TolSCl	87:13	87
2	p-ClC ₆ H ₄ SCl	85:15	90
3	2,4,6-(CH ₃) ₃ C ₆ H ₂ SCl	99:1	90

The effects of various factors, such as the nature of the Lewis acid, electronic and steric properties of the arylthio group, temperature and polarity of the solvent, have been examined in the reaction of adducts **2** with vinyl ether **4**. It is well known that stereoselectivity and sometimes even regioselectivity of electrophilic alkylations are sensitive to the Lewis acid used.¹¹ It has also been reported that the choice of Lewis acid may control the stereoselectivity of episulfonium ion reactions.¹² In the reaction studied, the Lewis acid is directly involved in the generation of an electrophile (which presumably has an episulfonium-like structure, *vide infra*) from adducts **2**. The structure of this electrophile and, consequently, reaction stereoselectivity and reactivity of the reactions of this intermediate may depend on the nature (e.g., acid strength, size, metal affinity to oxygen and sulfur) of the Lewis acid. We found, however, that the use of different Lewis acids does not lead to dramatic changes in the stereoselectivity of the glycosidation. In all cases, the reaction between **2** (Ar = *p*-Tol) and **4** proceeds with a high stereoselectivity with a predominant formation of the β -*gluco* isomer of **6a**. The best Lewis acid as judged by the stereoselectivity and combined yield of two isomers was ZnCl₂ (see Table 2).

One can speculate about the exact structure of the intermediate formed upon the action of a Lewis acid on the ArSCI adducts of D-glucal, but the involvement of ArS group in the stabilization of the species is obvious. The presence of different substituents on the aryl ring may change the bridging capacity of the sulfur and, therefore the intermediates with different structures could possibly alter the stereoselectivity of the reaction with vinyl ethers. Three different ArSCI have been used in the reaction: *p*-TolSCI, *p*-ClC₆H₄SCI, and 2,4,6-Me₃C₆H₂SCI (see Table 3). We found that differences in the electronic properties of the substituents on the aromatic ring do not significantly affect the stereoselectivity of the reaction. However, the use of the bulky 2,4,6-Me₃C₆H₂S group turned out to be very successful: only the β -*gluco* isomer of **6c** has been identified and isolated. These observations are in line with data for similar reactions of episulfonium ions.^{7a,13}

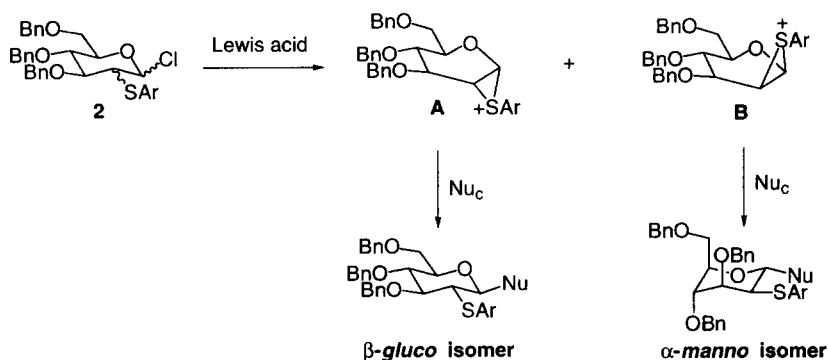
To determine the possible effect of temperature, the reaction of **2** (Ar = *p*-Tol) with **4** in the presence of SnCl₄ has been conducted at -78, -25, and 20 °C. In all three cases the stereoselectivity of glycosidation was the same (β -*gluco* : α -*manno* = 87:13).

We found that tri-*O*-acetyl-D-glucal (**21**) can also be employed instead of the benzyl protected analog, although the use of the former is much less practical. The addition of ArSCI to glucal **21** occurs at a sufficient rate only in CH₃CN which has to be removed before adding a Lewis acid and a nucleophile.¹⁴ The overall yield of *C*-glycoside **22** prepared by reacting vinyl ether **4** with ArSCI adducts of tri-*O*-acetyl-D-glucal was 51% (β -*gluco* : α -*manno* = 78:22).

The structures of all the synthesized *C*-glycosides were determined using ¹H NMR, ¹³C NMR (DEPT), ¹H-¹H homonuclear decoupling, and, in some cases, 2D ¹H-¹H homonuclear COSY and 2D ¹H-¹³C heteronuclear correlations. The composition of the compounds was confirmed by HRMS data and, in some cases, elemental analysis. The purity of the compounds (not less than 95%) has been estimated by NMR spectroscopy. The ratios of β -*gluco* and α -*manno* isomers were determined from ¹H NMR spectra. For compounds **6a**, **6b**, and **20** the minor isomers have been isolated and their α -*manno* configuration and conformation, as shown in Scheme 4, were unambiguously established upon analysis of the coupling constants in the ¹H NMR spectra. In the cases of **7**, **8**, **10-13**, and **18** the minor isomers have not been isolated and their

structure as α -manno derivatives was accepted by analogy. It is significant that, in contrast to 2-ArS substituted α -O-mannopyranosides,¹³ the corresponding C-glycosides have the conformation with axial substituents at C-3, C-4, and C-5 and an equatorial ArS group at C-2 and an equatorial alkyl chain at C-1 (see ¹H NMR data for α -manno-6a, α -manno-6b, and α -manno-22 in Experimental Section).

In all the reactions we performed, only two isomeric C-glucosides, β -gluco and α -manno, have been obtained. Both of them are the result of the *trans* addition of the ArS group and the carbon nucleophile across the double bond of the glucal. The *trans* configuration of the C-glycosides suggests that the intermediates formed upon the action of a Lewis acid on a mixture of the ArSCl adducts are likely to have the episulfonium ion structure. The comparison of the ratio of (α -gluco + β -gluco) : α -manno isomers for p-TolSCl adducts of D-glucal (88:12, CH₂Cl₂) with the ratio of two isomeric C-glycosides 6a (β -gluco : α -manno = 87:13, CH₂Cl₂) initially suggested that the stereoselectivity of glycosidation solely depends on the ratio of ArSCl adducts of D-glucal. However, when CH₃NO₂ was used in the reaction sequence, the ratio of TolSCl adducts (α -gluco : α -manno = 64:36, ¹H NMR data) did not coincide with the isomer ratio of 6a (β -gluco : α -manno = 49:51). Moreover, in the presence of SnCl₄ and in a solution of CH₃NO₂, vinyl ether 4 reacted with the mixture of adducts 2 ([α -gluco + β -gluco] : α -manno = 88:12) prepared in CH₂Cl₂ (the latter was removed before the reaction with compound 4) to give β -gluco and α -manno isomers of 6 in a ratio of 62:38. These data indicate that, if the reaction intermediate is the episulfonium ion, there is an interconversion between two forms of the intermediate, A and B (Scheme 4).^{7a} The equilibrium of two episulfonium ions is considered to be rather unique¹³ but it has been reported for a similar glucal-based episelenium species.¹⁵ One can suggest that the interconversion of intermediates A and B takes place through the formation of the glucal and the ArS⁺.^{7a} However, the glucal and the ArS⁺ would react with carbon nucleophiles but no products of these reactions have been detected. Another possible explanation of epimerization at C-2 would be the formation of the 2-arylsulfanyl substituted glucal as an intermediate. A recent theoretical study of 1-methoxy-2-sulfanylethan-1-yl cation does not exclude this possibility.¹⁶ However, the later mechanism cannot satisfactory explain the exclusive formation of the *trans* (β -gluco and α -manno) isomers.



Scheme 4.

It is noteworthy that the reaction studied requires special care to avoid traces of water. It is known that, under conditions when traces of water are not removed completely, episulfonium ions react with H_2O to give β -(arylsulfanyl)alkyl alcohols.¹² In the case of using vinyl ethers as nucleophiles, the presence of H_2O in the reaction mixture resulted in the formation of *O*-glycosides instead of *C*-glycosides. For example, the reaction of adducts **2** with methyl or ethyl vinyl ether in the presence of ZnCl_2 , which was exposed to the air for a few minutes, provided the α and β -anomers of methyl or ethyl 3,4,6-tri-*O*-benzyl-2-deoxy-2-(phenylsulfanyl)- β -D-glucopyranoside as the major product in yield of 73% and 49%, respectively, as well as up to 10% of 3,4,6-tri-*O*-benzyl-2-(*p*-tolylsulfanyl)-D-glucose. It seems apparent that the vinyl ethers reacted with H_2O to give the corresponding aldehydes and alcohols (MeOH or EtOH) and the alcohol formed then reacted with the episulfonium-like intermediate to yield the *O*-glycoside instead of the *C*-glycoside.

In summary, we have shown that the reaction of vinyl ethers with ArSCl adducts of D-glucal can be used for the highly stereoselective synthesis of β -C-glucosides. The reaction takes place through the formation of the five-membered sulfonium salt that can be quenched with H_2O , MeOH, or *n*- Bu_4NBH_4 to give an aldehyde, an acetal, or a vinyl ether, respectively.

Experimental Section

^1H and ^{13}C NMR spectra were recorded in CDCl_3 unless stated otherwise. Coupling constants, *J*, are reported in Hz. Preparative TLC were carried out by using glass plates, 200 x 200 mm, with an unfixed layer of Merck silica gel 60 (230–400 mesh). Analytical TLC were performed on Merck precoated 0.2 mm plates of silica gel 60 F₂₅₄. All reactions were carried out under an atmosphere of dry nitrogen using flame-dried glassware and freshly distilled and dried solvents.

Arylsulfonyl chlorides were obtained from the corresponding thiophenols by using SO_2Cl_2 .¹⁶ Methyl vinyl ether was synthesized from *n*-butyl vinyl ether and MeOH in the presence of $\text{Hg}(\text{OAc})_2$.¹⁷ 1-Methoxy-2-methyl-1-propene, 1-methoxy-1-cyclohexene, and 1-methoxy-1-cyclopropene were prepared by pyrolysis of 1,1-dimethoxy-2-methylpropane, 1,1-dimethoxycyclohexane, and 1,1-dimethoxycyclopentane, respectively, in the presence of *p*-toluenesulfonic acid.¹⁸ Ethylene glycol methyl vinyl ether was synthesized from ethylene glycol vinyl ether.¹⁹ Other chemicals are commercially available (Aldrich).

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-2-methylpropanal (β -gluco-6a) and 2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-2-methylpropanal (α -manno-6a). To a solution of 159 mg (1.0 mmol) *p*-TolSCl in 10 mL CH_2Cl_2 at room temperature was added 416 mg (1 mmol) 3,4,6-tri-*O*-benzyl-D-glucal (the color changed from yellow to colorless). After 10 min the mixture was cooled to -78°C and a solution of 103 mg (1.2 mmol) 2-methyl-1-methoxypropene in 2 mL CH_2Cl_2 was introduced. Then a solution of 0.14 mL (1.2 mmol) SnCl_4 in 2 mL CH_2Cl_2 was added dropwise. The mixture was stirred 30 min at -78°C , quenched with an aqueous saturated solution of NaHCO_3 , extracted with ether, and dried over Na_2SO_4 . Preparative TLC (ether - hexane, 1 : 4) of the crude material after solvent removal in vacuo afforded two isomers (β -gluco : α -manno = 87 : 13). Yield 530 mg (76 %). Data for β -gluco-6a. TLC: R_f (ether - hexane, 1 : 1) = 0.50. $[\alpha]^{21}_D = -52.2$ (c 2.46, CHCl_3).

^1H NMR (300 MHz): 0.90, 1.14 (two s, 6 H), 2.32 (s, 3 H), 3.07 (t, $J = 10.8$, 1 H), 3.48 (ddd, $J = 9.8$, 4.0, 2.5, 1 H), 3.60 (d, $J = 10.8$, 1 H), 3.67 (m, 2 H), 3.78 (dd, $J = 11.0$, 4.0, 1 H), 3.77 (dd, $J = 11.0$, 2.5, 1 H), 4.56, 4.62 (two d, $J = 12.5$, 2 H), 4.67, 4.89 (two d, $J = 10.5$, 2 H), 4.91, 5.05 (two d, $J = 10.5$, 2 H), 7.28 (m, 19 H), 9.72 (s, 1 H). ^{13}C NMR (75 MHz): 15.7, 21.1, 22.0, 50.1, 54.6, 69.2, 73.5, 75.1, 76.6, 79.7, 79.9, 92.2, 95.9, 127.6, 127.7, 127.8, 128.0, 128.3, 128.4, 128.5, 128.6, 130.0, 130.7, 131.6, 137.0, 138.4, 199.7. HRMS: calcd for $\text{C}_{38}\text{H}_{42}\text{SO}_5$ (M^+) m/e 610.2753, found m/e 610.2760. Data for α -manno-6a. TLC: R_f (ether - hexane, 1 : 1) = 0.56. $[\alpha]_D^{21} = +70.9^\circ$ (c 0.186, CHCl_3). ^1H NMR (300 MHz): 0.995, 1.08 (two s, 6 H), 2.185 (s, 3 H), 3.52 (dd, $J = 10.5$, 3.2, 1 H), 3.63 (dd, $J = 3.8$, 2.0, 1 H), 3.69 (dd, $J = 10.0$, 6.4, 1 H), 3.83 (dd, $J = 10.0$, 7.5, 1 H), 3.90 (br.t, $J = 3.2$, 1 H), 4.08 (d, $J = 10.5$, 1 H), 4.175 (br.t, $J = 6.8$, 1 H), 4.53 (m, 6 H), 7.30 (m, 19 H), 9.61 (s, 1 H). ^{13}C NMR (75 MHz): 16.0, 20.8, 21.2, 50.0, 50.2, 68.1, 71.9, 73.0, 73.8, 73.1, 73.2, 75.0, 78.3, 127.8, 127.8, 127.9, 128.1, 128.3, 128.5, 128.6, 129.6, 130.1, 131.1, 201.6. HRMS calcd for $\text{C}_{38}\text{H}_{42}\text{SO}_5$ (M^+) m/e 610.2753, found m/e 610.2756.

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-((*p*-chlorophenyl)sulfanyl)- β -D-glucopyranosyl)-2-methylpropanal (β -gluco-6b) and 2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-((*p*-chlorophenyl)sulfanyl)- α -D-mannopyranosyl)-2-methylpropanal (α -manno-6b). Both compounds were synthesized using $p\text{-ClC}_6\text{H}_4\text{SCl}$ by method described for 6a with a combined yield of 90% (β -gluco-6b : α -manno-6b = 85 : 15, ^1H NMR data). Data for β -gluco-6b. TLC: R_f (ether - hexane, 1 : 1) = 0.54. IR (neat): 1718 cm^{-1} . ^1H NMR (500 MHz): 0.86, 1.13 (two s, 6 H), 3.10 (br.t, $J = 10.7$, 1 H), 3.55 (ddd, $J = 10.0$, 4.0, 2.3, 1 H), 3.65 (d, $J = 10.7$, 1 H), 3.66 (m, 2 H), 3.72 (dd, $J = 11.0$, 2.3, 1 H), 3.74 (dd, $J = 11.0$, 4.0, 1 H), 4.62, 4.68 (two d, $J = 12.5$, 2 H), 4.75, 4.96 (two d, $J = 11.0$, 2 H), 5.00, 5.06 (two d, $J = 10.5$, 2 H), 7.35 (m, 19 H), 9.76 (s, 1 H). ^{13}C NMR (125 MHz): 15.4, 21.9, 50.0, 54.8, 68.9, 73.4, 75.1, 76.7, 79.6, 79.7, 81.7, 85.8, 127.5, 127.6, 127.6, 127.8, 127.8, 127.9, 128.2, 128.3, 128.4, 128.5, 129.1, 131.7, 133.0, 133.5, 138.0, 138.3, 199.3. HRMS calcd for $\text{C}_{37}\text{H}_{39}\text{SO}_5\text{Cl}$ (M^+) m/e 630.2207, found m/e 630.2227. Data for α -manno-6b. TLC: R_f (ether - hexane, 1 : 1) = 0.57. IR (neat): 1717 cm^{-1} . ^1H NMR (500 MHz): 0.955, 1.07 (two s, 6 H), 3.50 (dd, $J = 10.0$, 3.0, 1 H), 3.63 (dd, $J = 3.7$, 1.9, 1 H), 3.68 (dd, $J = 10.0$, 6.2, 1 H), 3.81 (dd, $J = 10.0$, 7.1, 1 H), 3.90 (br. t, $J = 3.7$, 3.0, 1 H), 4.07 (d, $J = 10.0$, 1 H), 4.175 (br.t, $J = 6.4$, 1 H), 4.52 (m, 6 H), 7.30 (m, 19 H). ^{13}C NMR (125 MHz): 15.8, 20.7, 49.9, 50.1, 67.9, 71.9, 73.1, 73.7, 72.7, 72.9, 74.8, 78.2, 127.4, 127.6, 127.7, 127.7, 127.8, 127.9, 128.0, 128.2, 128.4, 128.5, 129.3, 131.8, 133.1, 137.4, 137.9, 138.1, 201.3. HRMS calcd for $\text{C}_{37}\text{H}_{39}\text{SO}_5\text{Cl}$ (M^+) m/e 630.2207, found m/e 630.2240.

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(mesitylsulfanyl)- β -D-glucopyranosyl)-2-methylpropanal (β -gluco-6c). This derivative was synthesized by the method described for 6a with a yield of 90%. TLC: R_f (ether - hexane, 1 : 1) = 0.53. IR (neat): 1724 cm^{-1} . ^1H NMR (500 MHz): 1.17, 1.26 (two s, 6 H), 2.27 (s, 3 H), 2.49 (s, 6 H), 3.47 (dd, $J = 6.7$, 4.6, 1 H), 3.53 (dd, $J = 9.0$, 2.9, 1 H), 3.56 (br. t, $J = 4.6$, 2.9, 1 H), 3.63 (dd, $J = 11.0$, 5.4, 1 H), 3.68 (dd, $J = 11.0$, 2.4, 1 H), 3.76 (ddd, $J = 9.0$, 5.4, 2.4, 1 H), 3.84 (d, $J = 6.7$, 1 H), 4.10, 4.13 (two d, $J = 11.7$, 2 H), 4.20, 4.26 (two d, $J = 11.2$, 2 H), 4.58, 4.60 (two d, $J = 11.2$, 2 H), 6.87, 7.09, 7.28, 7.35 (four m, 17 H), 9.70 (s, 1 H). ^{13}C NMR (75 MHz): 18.5, 19.1, 21.0, 23.5, 45.0, 50.5, 70.1, 71.1, 72.3, 73.4, 78.0, 78.3, 80.4, 83.6, 127.4, 127.5, 127.6, 127.7, 127.8, 127.9, 128.0, 128.3, 128.4, 137.8, 137.9, 138.2, 138.4, 142.7, 203.3. HRMS calcd for $\text{C}_{40}\text{H}_{46}\text{SO}_5$ (M^+) m/e 638.3066, found m/e 638.3082.

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-1,1-dimethoxyethane (7) was prepared by the method described for **6a** but the reaction mixture was quenched with MeOH and K₂CO₃. Preparative TLC (ether - hexane, 1 : 4) afforded a mixture of β -*gluco* and α -*manno* isomers in a ratio of 87 : 13 in a combined yield of 76%. Data for β -*gluco*-7. TLC: R_f (hexane - ether, 1 : 1) = 0.55. ¹H NMR (C₆D₆, 250 MHz): 1.27, 1.52 (two s, 6 H), 1.98 (s, 3 H), 3.34, 3.39 (two s, 6 H), 3.60 (m, 6 H), 3.83 (d, *J* = 9.5, 1 H), 4.37, 4.43 (two d, *J* = 11.8, 2 H), 4.51, 4.77 (two d, *J* = 11.8, 2 H), 4.71, 4.95 (two d, *J* = 11.3, 2 H), 7.15 (m, 19 H). ¹³C NMR (C₆D₆, 50 MHz): 18.4, 19.0, 20.7, 45.1, 52.2, 57.4, 58.6, 70.1, 73.1, 73.7, 75.1, 78.7, 79.7, 82.9, 84.6, 110.9, 127.5, 127.6, 127.7, 127.9, 128.3, 129.1, 134.6, 136.2, 139.8. HRMS calcd for C₄₀H₄₈SO₆ (M⁺) m/e 656.3159, found m/e 656.3177.

Methyl 2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-2-methylpropyl Ether (8) was synthesized by analogy to **6a** using Bu₄NBH₄ (3 mmol) as a final nucleophile. After introducing Bu₄NBH₄, the mixture was stirred at -20 °C for 1 hour and poured into an aqueous saturated solution of NaHCO₃, extracted with ether, and dried over Na₂SO₄. Preparative TLC (ether - hexane, 1 : 4) of the crude material afforded a mixture of β -*gluco* and α -*manno* isomers (87:13) in a yield of 84%. Data for β -*gluco* -8. TLC: R_f (hexane - ether, 2 : 1) = 0.55. ¹H NMR (300 MHz): 1.02, 1.11 (two s, 6 H), 2.36 (s, 3 H), 3.385 (dd, *J* = 9.5, 8.5, 1 H), 3.69 (dd, *J* = 11.5, 2.5, 1 H), 3.465 (ddd, *J* = 9.5, 4.8, 2.5, 1 H), 3.61 (dd, *J* = 9.5, 7.0, 1 H), 3.71 (dd, *J* = 8.5, 7.0, 1H), 3.74 (dd, *J* = 4.8, 2.5, 1 H), 4.54, 4.72 (two d, *J* = 10.5, 2 H), 4.60, 4.65 (two d, *J* = 12.5, 2 H), 4.69, 4.87 (two d, *J* = 11.0, 2 H). ¹³C NMR (50 MHz): 21.1, 22.3, 22.7, 40.0, 51.1, 59.1, 69.9, 73.4, 73.9, 75.0, 80.6, 78.8, 79.5, 82.4, 84.2, 127.6, 127.8, 128.0, 128.3, 128.4, 128.5, 129.9, 131.0, 136.5, 138.4, 138.5. HRMS calcd for C₃₉H₄₆SO₅ (M⁺) m/e 626.3066, found 626.3058.

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)ethanal (10) was synthesized by the method described for **6a** using excess of methyl vinyl ether and 2 equiv. of SnCl₄. The preparative TLC (hexane - ether, 3:1) afforded a mixture of two isomers (β -*gluco* and α -*manno*) in a ratio of 10 : 1 (¹H NMR data) and combined yield of 63%. Data for β -*gluco* -10. TLC: R_f (hexane - ether, 1 : 1) = 0.45. [α]²¹_D = -17.14° (c 0.7, CHCl₃). IR (CHCl₃): 1726. ¹H NMR (300 MHz): 2.31 (s, 3 H), 2.59 (ddd, *J* = 12.7, 9.2, 2.7, 1 H), 2.94 (t, *J* = 10.4, 1 H), 3.12 (ddd, *J* = 12.7, 3, 1.5, 1 H), 3.50 (m, 5 H), 3.90 (ddd, *J* = 10.4, 9.2, 3, 1 H), 4.47, 4.57 (two d, *J* = 12.3, 2 H), 4.57, 4.86 (two d, *J* = 10.8, 2 H), 4.89, 5.07 (two d, *J* = 10.4, 2 H), 7.27 (m, 19 H), 9.75 (dd, *J* = 2.7, 1.5, 1 H). ¹³C NMR (75 MHz): 21.1, 41.8, 46.9, 56.9, 73.7, 75.1, 76.3, 76.5, 79.1, 79.6, 84.5, 127.6, 127.8, 127.7, 128.5, 130.0, 132.4, 137.7, 138.0, 138.1, 138.4, 184.9. HRMS calcd for C₃₆H₃₈SO₅ (M⁺) m/e 582.2430, found m/e 582.2423.

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-1,1-dimethoxyethane (11). The compound was synthesized by the method described for **6a** using methyl vinyl ether (ca. 3 mmol) as a carbon nucleophile and MeOH as a final quencher. The combined yield of two isomers (β -*gluco* and α -*manno*) in a ratio of 91 : 9, ¹H NMR data) is 34%. Data for β -*gluco*-11. R_f (ether-hexane, 1:1) = 0.42. [α]²¹_D = -31.1° (c 0.74, CHCl₃). ¹H NMR (300 MHz): 1.62 (m, 1H), 2.32 (s, 3H), 2.60 (m, 1H), 2.93 (t, *J* = 10.6, 1H), 3.33 (s, 3H), 3.37 (s, 3H), 3.41 (ddd, *J* = 9.1, 2.9, 1.9, 1H), 3.48 (ddd, *J* = 10.6, 2.9, 1.9, 1H), 3.57 (dd, *J* = 10.6, 8.7, 1H), 3.67 (dd, *J* = 9.1, 8.7, 1H), 3.72 (d, 2H), 4.70 (dd, *J* = 8.9, 2.8, 1H), 4.55, 4.63 (two d, *J* = 12.3, 2H), 4.62, 4.86 (two d, *J* = 10.8, 2H), 4.90, 5.08 (two d, *J* = 19.3, 2H), 7.25 (m, 19H). ¹³C NMR (75

MHz): 21.0, 35.8, 52.3, 53.3, 57.3, 68.3, 73.3, 74.3, 76.1, 77.2, 78.5, 79.7, 84.8, 102.0, 127.3, 127.6, 127.8, 128.3, 128.4, 129.7, 129.7, 131.6, 132.2, 137.1, 138.1, 138.2, 138.4. HRMS calcd for $C_{38}H_{44}SO_6$ (M^+) m/e 628.2859, found m/e 628.2854.

Methyl 2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)ethyl Ether (12).

The compound was synthesized by the method described for **6a** using methyl vinyl ether (ca. 3 mmol) as a carbon nucleophile and *n*-Bu₄NBH₄ as a final quencher. A combined yield of two isomers (β -*gluco* and α -*manno* in a ratio of 91 : 9, ¹H NMR data) is 23%. Data for β -*gluco*-12. R_f (ether-hexane, 1:1) = 0.44. $[\alpha]^{21}_D = -53.0^\circ$ (c 0.42, CHCl₃). ¹H NMR (300 MHz): 1.62 (m, 1H), 2.30 (s, 3H), 2.51 (m, 1H), 2.89 (t, $J = 10.4$, 1H), 3.30 (s, 3H), 3.36 (ddd, $J = 8.9$, 3.4, 3.2, 1H), 3.44 (m, 2H), 3.53 (t, $J = 10.4$, 8.9, 1H), 3.54 (t, 1H), 3.63 (t, $J = 8.9$, 1H), 3.70 (m, 2H), 4.53, 4.60 (two d, $J = 12.2$, 2H), 4.58, 4.84 (two d, $J = 10.8$, 2H), 4.88, 5.06 (two d, $J = 10.3$, 2H), 7.28 (m, 19H). ¹³C NMR (75 MHz): 21.0, 32.6, 57.3, 58.5, 69.0, 69.2, 73.4, 74.9, 76.1, 76.6, 78.6, 79.8, 84.8, 127.5, 127.6, 127.7, 127.8, 128.0, 128.3, 128.4, 129.7, 131.6, 132.3, 137.1, 138.2, 138.5. HRMS calcd for $C_{37}H_{42}SO_5$ (M^+) m/e 598.2753, found m/e 598.2747.

Methyl 1-Deuterio-2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-2-methylpropyl Ether (13). The ether was formed as a mixture of two isomers (β -*gluco* : α -*manno* = 92:8) in 59% combined yield using *n*-Bu₄NBD₄ as a final nucleophile. Data for pure β -*gluco* isomer. TLC: R_f (ether-hexane, 1:2) = 0.52. ¹H NMR (300 MHz): 1.00 (s, 3H), 1.09 (s, 3H), 2.33 (s, 3H), 3.30 (s, 3H), 3.36 (dd, $J = 9.4$, 7.9, 1H), 3.38 (dd, $J = 13.8$, 4.3, 1H), 3.44 (ddd, $J = 9.2$, 4.3, 2.6, 1H), 3.51 (d, $J = 9.4$, 1H), 3.57 (dd, $J = 9.2$, 6.8, 1H), 3.68 (m, 3H), 4.52, 4.84 (two d, $J = 10.8$, 2H), 4.56, 4.62 (two d, $J = 12.2$, 2H), 4.65, 4.70 (two d, $J = 11.1$, 2H), 7.25 (m, 19H). ¹³C NMR (75 MHz): 21.3, 22.4, 22.5, 22.8, 51.2, 59.1, 70.1, 73.5, 74.1, 75.2, 80.3, 78.9, 79.6, 82.5, 84.3, 127.6, 127.7, 127.9, 128.1, 128.3, 128.4, 128.5, 128.6, 129.9, 131.1, 136.5, 138.5, 138.6, 138.9. HRMS calcd $C_{39}H_{45}DO_5SNa$ ($M^+ + Na^+$) m/e 650.3026, found ($M^+ + Na^+$) 650.3043.

[2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)-1-(oxoethyl)]trimethylsilane (18) was synthesized by the method described for **6a** using (1-methoxyvinyl)trimethylsilane as a carbon nucleophile. The preparative TLC (hexane-ether, 3:1) afforded a mixture of two isomers (β -*gluco* and α -*manno*;) in a ratio of 95:5 and a combined yield of 59%. Data for β -*gluco*-18. TLC: R_f (hexane - ether, 1 : 1) = 0.60. $[\alpha]^{22}_D = -15.9^\circ$ (c 0.22, CHCl₃). ¹H NMR (300 MHz): 0.21 (s, 9H), 2.97 (dd, $J = 16.1$, 8.5, 1H), 2.98 (t, $J = 10.2$, 1H), 3.22 (dd, $J = 16.1$, 2.7, 1H), 3.38 (ddd, $J = 9.7$, 1.9, 3.4, 1H), 3.57 (dd, $J = 10.0$, 3.4, 1H), 3.58 (dd, $J = 10.2$, 9.0, 1H), 3.67 (dd, $J = 9.7$, 9.0, 1H), 3.72 (dd, $J = 10.0$, 1.9, 1H), 3.98 (ddd, $J = 10.2$, 8.5, 2.7, 1H), 4.51, 4.61 (two d, $J = 12.2$), 4.63, 4.87 (two d, $J = 10.7$), 4.93, 5.08 (two d, $J = 10.4$), 7.28 (m, 19H). ¹³C NMR (75 MHz): -3.3, 21.1, 51.0, 56.5, 68.8, 73.4, 74.9, 75.8, 75.9, 76.7, 79.6, 84.5, 127.6, 129.8, 132.3, 137.3, 138.1, 138.2, 138.5, 246.3. HRMS calcd $C_{39}H_{47}SiO_5S$ (MH^+) m/e 655.2915, found (MH^+) 655.2922.

2-(3,4,6-Tri-*O*-benzyl-2-deoxy-2-(*p*-tolylsulfanyl)- β -D-glucopyranosyl)cyclohexanone (20) was synthesized by the method described for **6a** using 1-methoxy-1-cyclohexene as a carbon nucleophile. The product was obtained as a mixture of two diastereomers (both are β -*gluco* isomers) in a ratio of 10 : 1 and a combined yield of 61%. Data for the major isomer (with (S)-configuration at the chiral center in the lateral chain). Mp 104-105 °C. TLC: R_f (hexane - ether, 1 : 1) = 0.43. $[\alpha]^{21}_D = -53.9^\circ$ (c 1.19, CHCl₃). IR

(CHCl₃): 1714 cm⁻¹. ¹H NMR (500 MHz): 1.65, 2.30 (two m, 6 H), 2.33 (s, 3 H), 3.12 (b.t, *J* = 11.3, 9.6, 1 H), 3.10 (b.dd, *J* = 12.1, 6.0, 1 H), 3.47 (m, *J* = 9.6, 3.3, 1, 1 H), 3.61 (t, *J* = 9.6, 1 H), 3.69 (t, *J* = 9.6, 9.5, 1 H), 3.70 (dd, *J* = 11.0, 1, 1 H), 3.785 (dd, *J* = 11.0, 3.3, 1 H), 4.11 (dd, *J* = 11.3, 1.0, 1 H), 4.51, 4.60 (two d, *J* = 12.1, 2 H), 4.69, 4.86 (two d, *J* = 10.7, 2 H), 4.93, 5.06 (two d, *J* = 10.3, 2 H), 7.30 (m, 19 H). ¹³C NMR (50 MHz): 21.0, 24.1, 24.7, 25.7, 41.8, 50.2, 53.7, 68.8, 73.1, 74.8, 76.1, 76.7, 78.8, 79.8, 84.5, 127.3, 127.5, 127.55, 127.6, 127.8, 127.9, 128.0, 128.2, 128.3, 129.7, 129.8, 131.0, 132.7, 137.5, 138.3, 138.5, 209.8. HRMS calcd for C₄₀H₄₄SO₅ (M⁺) *m/e* 636.2898, found for *m/e* 636.2923. Anal. Calcd for C₄₀H₄₄SO₅: C, 75.44; H, 6.96; S, 5.03. Found: C, 75.38, H, 6.93; S, 4.97. Data for the minor isomer. R_f (hexane-ether, 1:1) = 0.50. ¹H NMR (500 MHz): 1.6 (m, 6 H), 2.28 (s, 3 H), 3.07 (b.t, *J* = 8.5, 1 H), 3.36 (m, *J* = 9.8, 5.2, 2.1, 1 H), 3.39 (dd, *J* = 10.7, 1.8, 1 H), 3.52 (dd, *J* = 10.7, 8.9, 1 H), 3.62 (b.t, *J* = 10.7, 8.9, 1 H), 3.66 (dd, *J* = 11.0, 5.2, 1 H), 3.71 (dd, *J* = 11.0, 2.1, 1 H), 3.805 (b.t, *J* = 10.7, 9.8, 1 H), 4.52, 4.58 (two d, *J* = 12.1, 2 H), 4.58, 4.82 (two d, *J* = 10.9, 2 H), 4.84, 5.00 (two d, *J* = 10.4, 2 H), 7.30 (m, 19 H). ¹³C NMR (75 MHz): 20.9, 21.3, 21.6, 24.3, 42.2, 49.9, 52.9, 69.2, 73.1, 74.9, 75.7, 79.5, 79.6, 81.2, 85.3, 127.3, 127.6, 127.75, 127.8, 128.0, 128.1, 128.2, 128.2, 128.3, 129.6, 129.8, 131.2, 131.5, 132.7, 136.6, 138.1, 138.4, 209.5.

2-(3,4,6-Tri-*O*-acetyl-2-deoxy-2-(*p*-tolylsulfanyl)-β-D-glucopyranosyl)-2-methylpropanal (β-*gluco*-22) and 2-(3,4,6-Tri-*O*-acetyl-2-deoxy-2-(*p*-tolylsulfanyl)-α-D-mannopyranosyl)-2-methylpropanal (α-*manno*-22). To a solution of 159 mg (1.00 mmol) *p*-TolSCl in 3 mL CH₃CN at 0 °C was added 272 mg (1.00 mmol) 3,4,6-tri-*O*-acetyl-D-glucal (**21**) (the color changed from yellow to colorless). After 10 min the solvent was evaporated and the residue was dissolved in 20 mL CH₂Cl₂. This solution was cooled to -78 °C and a solution of 103 mg (1.2 mmol) 1-methoxy-2-methyl-1-propene in 2 mL CH₂Cl₂ was introduced. After that a solution of 0.14 mL (1.2 mmol) SnCl₄ in 2 mL CH₂Cl₂ was added dropwise. The mixture was stirred 30 min at 0 °C, quenched with a saturated solution of NaHCO₃ in water, extracted with ether, and dried over Na₂SO₄. Preparative TLC (ether - hexane, 1:2) of the crude material after solvent removal in vacuo afforded **22** as a mixture of β-*gluco* and α-*manno* isomers in a ratio of 78:22 and a combined yield of 51%. Data for β-*gluco*-22. TLC: R_f (hexane - ether, 1:1) = 0.25. IR (neat): 1748 cm⁻¹. ¹H NMR (250 MHz): 0.96, 1.11 (two s, 6 H), 1.92, 2.03, 2.07 (three s, 9 H), 2.31 (s, 3 H), 3.15 (t, *J* = 10.8, 1 H), 3.66 (d, *J* = 10.8, 1 H), 3.68 (m, *J* = 9.5, 5.3, 2.6, 1 H), 4.12 (dd, *J* = 12.2, 2.6, 1 H), 4.22 (dd, *J* = 12.2, 5.3, 1 H), 4.97 (t, *J* = 9.5, 1 H), 5.23 (dd, *J* = 10.8, 9.5, 1 H), 7.16 (m, 4 H), 9.66 (s, 1 H). ¹³C NMR (62.5 MHz): 15.3, 20.6, 20.6, 20.7, 20.9, 21.5, 49.9, 51.9, 62.2, 69.7, 75.5, 75.7, 81.3, 129.9, 130.5, 132.6, 137.4, 169.4, 169.8, 170.5, 199.2. HRMS calcd for C₂₃H₃₀O₈S (M⁺) *m/e* 466.1661, found *m/e* 466.1673. Data for α-*manno*-22. TLC: R_f (hexane - ether, 1:1) = 0.28. IR (neat): 1745 cm⁻¹. ¹H NMR (500 MHz): 1.11, 1.13 (two s, 6 H), 2.055, 2.06, 2.08 (3 s, 9 H), 2.32 (s, 3 H), 3.43 (dd, *J* = 9.5, 3.7, 1 H), 4.08 (d, *J* = 9.5, 1 H), 4.12 (m, 2 H), 4.75 (m, *J* = 11.6, 4.0, 1.5, 1 H), 4.88 (dd, *J* = 4.9, 1.5, 1 H), 5.22 (br.t, *J* = 4.9, 3.7, 1 H), 7.19 (m, 4 H), 9.61 (s, 1 H). ¹³C NMR (125 MHz): 16.4, 19.5, 20.6, 20.7, 20.8, 21.1, 48.0, 50.1, 60.1, 67.7, 69.8, 72.6, 74.2, 130.0, 132.6, 138.4, 170.4, 169.4, 201.5. HRMS calcd for C₂₃H₃₀O₈S (M⁺) *m/e* 466.1661, found *m/e* 466.1671.

Crystallographic Data Sulfone of the Major Isomer of 20. The compound was crystallized by dissolving in benzene and then layering with hexane. Colorless plate crystals were grown of diffraction quality. Data was collected using a Siemens SMART CCD (charge coupled device) based diffractometer

equipped with an LT-2 low-temperature apparatus operating at 213 °K. A suitable crystal was chosen and mounted on a glass fiber using grease. Data were measured using omega scans of 0.3 ° per frame for 30 seconds, such that a hemisphere was collected. A total of 1271 frames were collected with a final resolution of 0.90 Å. The first 50 frames were recollected at the end of data collection to monitor for decay. Cell parameters were retrieved using SMART²⁰ software and refined using SAINT on all observed reflections. Data reduction was performed using the SAINT software²¹ which corrects for Lp and decay. Absorption corrections were applied using SADABS.²² The structures are solved by the direct method using the SHELXS-90 program²³ and refined by least squares method on F², SHELXL-97,²⁴ incorporated in SHELXTL-PC V 5.03.²⁵ The structure was solved in the space group P1 (#1) by analysis of systematic absences. All non-hydrogen atoms are refined anisotropically. The Flack parameter, 0.1(3) and refinement of the other enantiomer suggested the reported conformation. The crystal used for the diffraction study showed no decomposition during data collection.

Acknowledgment. Authors acknowledge ND EPSCoR (grant No. OSR-9452892) and the National Institute of General Medical Sciences (grant No. 1 R15 GM/OD55965-01) for financial support. We would like to thank Dr. R. Staples (Harvard University) for the X-ray crystallographic determination.

References and Notes

1. a) Levy, D. E.; Tang, C. *The Chemistry of C-Glycosides*. Tetrahedron Organic Chemistry Series, Volume 13; Elsevier Science Ltd.: Tarrytown, NY, 1995. b) Postema, M. H. D. *C-Glycoside Synthesis*; CRC Press Inc.: Boca Raton, FL, 1995.
2. a) Lown, J. W. *Chem. Soc. Rev.* **1993**, 165. b) Rohr, J.; Thiericke, R. *Nat. Prod. Rep.* **1992**, 103. c) Murenetz, N. V. *Antibiot. Khimioter.* **1990**, 35, 47. d) Hacksell, U.; Daves, G. D., Jr. *Prog. Med. Chem.* **1985**, 22, 1.
3. Bertozzi, C. and Bednarski, M. Synthesis of C-Glycosides; Stable Mimics of O-Glycosidic Linkages. In: *Modern Methods in Carbohydrate Synthesis*; Khan, S. H. and O'Neill, R. A., Eds.; Hardwood Academic Publishers: Amsterdam, 1996.
4. Stick, R.V. *Topics in Current Chemistry* **1997**, 187, 187.
5. Stereoselective synthesis of β-C-glycosides: Evans, D. A.; Trotter, B. W.; Cote, B. *Tetrahedron Lett.* **1998**, 39, 1709. b) Minehan, T. G.; Kishi, Y. *Tetrahedron Lett.* **1997**, 38, 1709. c) Best, W. M.; Ferro, V.; Harle, J.; Stick, R. V.; Tilbrook, D. M. *Aust. J. Chem.* **1997**, 50, 463. d) Roe, B. A.; Boojamra, C. G.; Griggs, J. L.; Bertozzi, C. R. *J. Org. Chem.* **1996**, 61, 6442. e) Ayadi, E.; Czernecki, S.; Xie, J. *Chem. Commun.* **1996**, 347. f) Wei, A.; Haudrechy, A.; Audin, C.; Jin, H.-C.; Haundrechy-Bretel, N.; Kishi, Y. *J. Org. Chem.* **1995**, 60, 2160. g) Kawakami, H.; Ebata, T.; Koseki, K.; Okano, K.; Matsumoto, K.; Matsushita, H. *Heterocycles* **1993**, 93, 2765. h) Kobertz, W. A.; Bertozzi, C. R.; Bednarski, M. D. *Tetrahedron Lett.* **1992**, 33, 737. i) Stork, G.; Suh, H. S.; Kim, G. *J. Am. Chem. Soc.* **1991**, 113, 7054. j) Bellosta, V.; Czernecki, S. *J. Chem. Soc., Chem. Commun.* **1989**, 199. k) Lewis, M. D.; Cha, J. K.; Kishi, Y. *J. Am. Chem. Soc.* **1982**, 104, 4976.
6. Some recent publications on synthesis of C-glycosides: a) Christopher, J. A.; Kocienski, P. J.; Procter, M. *Synlett* **1998**, 425. b) Jegou, A.; Pacheco, C.; Veyrieres, A. *Synlett* **1998**, 81. c) Roche, D.; Banteli, R.; Winkler, T.; Casset, F.; Ernst, B. *Tetrahedron Lett.* **1998**, 39, 2545. d) Burkhardt, F.; Kessler, H. *Tetrahedron Lett.* **1998**, 39, 255. e) Moineau, C.; Bolitt, V.; Sinou, D. *J. Org. Chem.* **1998**, 63, 582. f) Csuk, R.; Kuhn, M.; Strohl, D. *Tetrahedron* **1997**, 53, 1311. g) Czernecki, S.; Ayadi, E.; Xie, J. *Tetrahedron Lett.* **1996**, 37, 9193. h) Casero, F.; Cipolla, L.; Lay, L.; Nicotra, F.; Panza, L.; Russo, G. J.

- Org. Chem.* **1996**, *61*, 3428. i) Dorgan, B. J.; Jackson, R. F. W. *Synlett* **1996**, 859. j) Thorn, S. N.; Gallagher, T. *Synlett* **1996**, 185.
7. a) Smoliakova, I. P.; Caple, R.; Gregory, D.; Smit, W. A.; Shashkov, A. S.; Chizhov, O. S. *J. Org. Chem.* **1995**, *60*, 1221. b) Smoliakova, I. P.; Caple, R.; Brenny, J. W.; Smit, W. A.; Kryschenko, Y. K.; Shashkov, A. S.; Chizov, O. S.; Krimer, M. Z.; Morar, G. V.; Kalyan, Y. B. *Synlett* **1995**, 275. c) Smoliakova, I. P.; Kim, Y. H.; Barnes, M. J.; Caple, R.; Smit, W. A.; Shashkov, A. S. *Mendeleev Commun.* **1995**, 15.
8. a) Smoliakova, I. P.; Smit, W. A.; Lutzenko, A. I. *Izv. Acad. Nauk SSSR, Ser. Khim.* **1987**, 119; *Chem. Abstr.* **1987**, *107*, 175155j. b) Smoliakova, I. P.; Smit, W. A. *Izv. Acad. Nauk SSSR, Ser. Khim.* **1988**, 2792; *Chem. Abstr.* **1989**, *110*, 231204a. c) Smoliakova, I. P.; Smit, W. A. *Izv. Acad. Nauk SSSR, Ser. Khim.* **1990**, 2036. *Chem. Abstr.* **1990**, *112*, 118356e. d) Smoliakova, I. P.; Smit, W. A.; Bubnov, Yu. N.; Shashkov, A. S. *Izv. Acad. Nauk SSSR, Ser. Khim.* **1991**, 1100; *Chem. Abstr.* **1991**, *115*, 158890t. e) Smoliakova, I. P.; Caple, R.; Magnuson, V. R.; Polyakov, V. R.; Smit, W. A.; Shashkov, A. S. *J. Chem. Soc. Perkin Trans. I* **1995**, 1065. f) Hayford, A.; Lovdahl, M.; Lazareva, M. I.; Kryschenko, Y. K.; Johnson, T.; Dilman, A. D.; Smoliakova, I. P.; Caple, R.; Smit, W. A. *Mendeleev Commun.* **1997**, 48. g) Lazareva, M. I.; Kryschenko, Y. K.; Caple, R.; Wakefield, D.; Hayford, A.; Smit, W. A.; Shashkov, A. S. *Tetrahedron Lett.* **1998**, *39*, 8787.
9. Another fact supporting the formation of the five-membered sulfonium salt is the result of the reaction of adducts **2** with 1-methoxybutadiene. In this case, the formation of the cyclic intermediate is impossible. Compound **23** was isolated in a low yield of 3%, other products were oligomers **24** (ca. 40%).



10. Our preliminary results on the synthesis of C-glycosides using 1-methoxy-2-methyl-1-propene have been described in our review, reference 12.
11. Santelli, M.; Pons, J.-M. *Lewis Acids and Selectivity in Organic Synthesis*, CRC Press: Boca Raton, FL, 1996.
12. Smit, W. A.; Caple, R.; Smoliakova, I. P. *Chem. Rev.* **1994**, *94*, 2359.
13. Grewal, G.; Kaila, N.; Franck, R. W. *J. Org. Chem.* **1992**, *57*, 2084.
14. Gordon, D. M.; Danishefsky, S. J. *J. Org. Chem.* **1991**, *56*, 3713.
15. Ito, Y.; Ogawa, T. *Tetrahedron* **1990**, *46*, 89.
16. Dudley, T. J.; Smoliakova, I. P.; Hoffmann, M. R. *J. Org. Chem.* **1999**, *64*, accepted for publication.
17. Fieser, M.; Fieser, L.F.P. *Reagents for Organic Synthesis*, Wiley: New York, 1975; Vol. 5, p 523.
18. Burgstahler, A. W.; Nordin, Z. C. *J. Am. Chem. Soc.* **1961**, *83*, 198.
19. Dukstra, H. B. *J. Am. Chem. Soc.* **1935**, *57*, 2257.
20. Merz, A. *Angew. Chem., Int. Ed. Engl.* **1973**, *12*, 846.
21. SMART V 4.043 Software for the CCD Detector System, Siemens Analytical Instruments Division, Madison, WI, 1995.
22. SAINT V 4.035 Software for the CCD Detector System, Siemens Analytical Instruments Division, Madison, WI, 1995.
23. SADABS. Program for Absorption Corrections Using Siemens CCD Based on the Method of B. Blessing. *Acta Cryst.* **1995**, *A51*, 33.
24. Sheldrick, G. M. SHELXS-90, Program for the Solution of Crystal Structure, University of Gottingen, Germany, 1986.
25. Sheldrick, G. M. SHELXL-97, Program for the Solution of Crystal Structure, University of Gottingen, Germany, 1997.
26. SHEXTL 5.03 (PC-Version), Program Library for Structure Solution and Molecular Graphics, Siemens Analytical Instruments Division, Madison, WI, 1995.